Neel Shah

Software and Data Systems · Python · Reliable Delivery

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SUMMARY

I build and run working software systems end to end, and I write down plainly what works and what does not. My main project, CourtVision, is a forecasting system for four sports with a web API, background services under a **watchdog**, a large automated test suite and a public evidence record. My own checks caught six of my earlier results as overstated, and I retracted them in public. Before that I worked in data and reporting roles at Sunsolor, DataAnnotation.Tech and Fortrex Securities.

PROFESSIONAL EXPERIENCE

Data Analyst Intern, Sunsolor · *Tempe, AZ · 2025*
Built live **Tableau / Power BI** suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in **Python**; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

Data Annotation Specialist, DataAnnotation.Tech · *Remote · 2024–2025*
Labeled **10,000+** structured samples for production ML training & LLM fine-tuning; authored corner-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst, Fortrex Securities · *Madison, WI · 2023–2024*
Built secure executive reporting interfaces (**99.9% uptime**); refined backend payment APIs & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

SELECTED PROJECT: COURTVISION

Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.
Evidence: github.com/neeljshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neeljshah/court-vision/blob/master/EVIDENCE.md

What I built

- **Web API and services:** built and run a **Python** web API (**FastAPI**, about 99 endpoints across 12 routers) and 9 long-running background services supervised by a **watchdog**.
- **Testing:** about **7,400 automated tests** (pytest), CI/CD on GitHub Actions, and 5 **Docker** images.
- **Outside benchmark:** scored every forecast against the market's closing price in four sports; on moneylines it lands near the close but never beats it, and I say so.
- **Published results:** pregame probabilities match or trail the devigged close and the in-game model is behind the live market, with n and 95% intervals; six withdrawn figures published with reasons at /evidence/retractions/; 16 findings on one template with limitations and verdict history.
- **Risk controls:** the trading side is **paper only**, behind six pre-registered go-live gates; no real order has been placed.
- **AI tooling:** built most of the system by directing **AI coding agents** (a planner model guiding cheaper executor models under strict ship gates).
- **Evidence site:** one unified, CI-gated site at https://neeljshah.github.io/court-vision/, built by one person directing **AI coding agents**; performance-budget and accessibility (axe) gates are hard CI steps with zero serious or critical violations; a PAPER ONLY desk record with receipts.
- **Video:** earlier work on the same project turned broadcast basketball video into player court positions on one consumer GPU.

EDUCATION & CERTIFICATIONS

University of Iowa: three years of data-science coursework. Coursework: math stats, probability theory, machine learning, financial analytics.

Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

Languages: Python (expert), SQL, JavaScript/TypeScript (working)

Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau, Power BI, dashboards

ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels

Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration